Soft Actor–Critic Based 3-D Deployment and Power Allocation in Cell-Free Unmanned Aerial Vehicle Networks

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***Abstract*—This letter investigates a cell-free unmanned aerial vehicles (UAV) network where UAVs serve as flying access points (FAPs) to overcome the inter-cell interference in conventional cell-based UAV networks and offer more uniform service. Considering the fairness among users, we aim to maximize the minimum user rate by joint optimizing three-dimensional (3D) deployment and power allocation of FAPs. To solve this complex high-dimentional non-convex problem, we firstly formulate it as a constrained Markov Decision Process (CMDP) guaranteeing the safety distance between FAPs. Then a soft actor-critic (SAC) based deep reinforcement learning (DRL) algorithm is proposed which adds maximum entropy term to the objective function. Simulation results demonstrate that the cell-free UAV network can achieve a higher minimum user rate than cell based UAV network, and SAC-based algorithm outperforms conventional DRL-based algorithms in terms of stability and long-term rewards.**

***Index Terms*—Cell-free unmanned aerial vehicle (UAV) networks, deep reinforcement learning (DRL), soft actor-critic (SAC), 3D deployment, power allocation.**

I. Introduction

IN CURRENT cell-based networks, each user connects to one access point (AP) that provides the strongest signal. However, this method cannot fulfill rapidly growing demands for data throughput and increasing wireless applications. Cell-free network is proposed as a promising framework, where numerous geographically distributed APs jointly serve the users with more uniform quality of service [1]. Considering the disadvantages of the terrestrial network in terms of limited coverage area, high deployment cost, and poor flexibility, cell-free unmanned aerial vehicle (UAV) network with flying APs (FAPs) has attracted a widespread attention [2], [3]. To fully utilize the mobility of UAVs to provide on demand services, resource allocation for cell-free UAV networks urgently needs to be investigated.

The work in [4] optimized trajectories of the FAPs to efficiently maximize coverage range in context of the vehicular network. In addition to trajectory design, power allocation and deployment also play a very important part in resource allocation of cell-free UAV networks. In [5], the problem of two-dimensional (2D) deployment of FAPs was solved through the combination of gradient-based and Gibbs sampling techniques. In [6], the authors investigated a 3D deployment of FAPs to maximize the downlink sum rate, where the optimal altitude and horizontal coordinates were obtained. In [7], a power allocation algorithm in the cell-free UAV network was studied to maximize the minimum signal to interference plus noise ratio (SINR) of users in uplink transmission, which ensured service fairness among users. Although the abovementioned works have been devoted to improving the performance of cell-free UAV networks, the deployment and power allocation were separately optimized with traditional convex optimization. To further develop the capability of the cell-free UAV network, it is necessary to jointly optimize deployment and power allocation. However, the problem involves both continuous and discrete variables, with high optimization dimensionality, making it extremely complex to solve using conventional optimization methods. Deep reinforcement learning (DRL) could be an efficient solution to tackle such an issue. In [8], a deep deterministic policy gradient (DDPG)-based approach was proposed to derive the joint 3D deployment and power allocation of single UAV base station. Furthermore, in [9], [10], the authors studied the joint optimization problem in multi-UAVs system, and a multi-agent Q-learning-based algorithm was proposed. Nevertheless, the above work only investigated the traditional cell-based UAV networks, without considering the cooperation between UAVs to provide uniform service.

For the first time, in this letter, we consider the joint 3D deployment and power allocation of FAPs in the cell-free UAV network, where all FAPs simultaneously serve all users in the same time-frequency resource. Specifically, to ensure fair service provision among users, we first formulate the joint optimization problem with the objective of maximizing the minimum user rate. Concerning the practical constraint in the cell-free UAV network, a minimum safety distance is guaranteed, which prevents collisions between FAPs. To tackle the complex problem, we propose a soft actor-critic (SAC)-based algorithm, which aims to maximize expected rewards while also maximizing entropy, and making it more conducive to finding the global optimal solution. Simulation results show the superior performance of our proposed algorithm in terms of stability and achievable minimum user rate.

II. System Model

As illustrated in Fig. 1, we consider a cell-free UAV network, which consists of M FAPs denoted by $\mathcal{M} = \{1, 2, \ldots, m, \ldots, M\}$, K users denoted by

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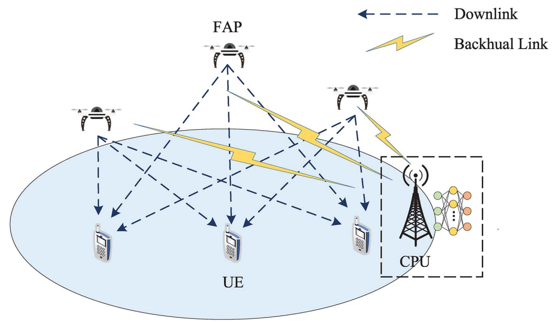

Fig. 1. Cell-free UAV networks.

$\mathcal{K} = \{1, 2, \ldots, k, \ldots, K\}$, and a base station. All FAPs and users are equipped with a single antenna. The base station acts as the central processing unit to provide wireless backhaul links to the FAPs. To support massive access with limited spectrum resources, we assume that all M FAPs simultaneously serve all K user equipments (UEs) in the same time-frequency resource via time-division duplex operation. Compared with traditional cell-based UAV networks, cell-free UAV networks can better coordinate interference, improve spectrum efficiency, and ensure fairness of communication between users. Let us denote the 3D position of the mth FAP by $\mathbf{q}_m = [x_m, y_m, z_m]$, and horizontal coordinates of kth user is denoted by $\mathbf{u}_k = [x_k, y_k]$.

A. Channel Model

The channel coefficient between kth user and mth FAP is denoted by $g_{m,k}$, which is modeled as

$$g_{m,k} = L_{m,k}^{1/2} h_{m,k}, \tag{1}$$

where $L_{m,k}$ represents the large-scale fading coefficient, and $h_{m,k} \sim \mathcal{CN}(0, 1)$ is the small-scale fading coefficient, which follows complex Gaussian distribution [11]. According to [12], the dB form of $L_{m,k}$ between the FAP and the ground user can be modelled by a probabilistic pathloss model, and the pathloss follows

$$\tilde{L}_{m,k} = 20 \log \left(\frac{4\pi f_c d_{m,k}}{c} \right) + P_{\mathrm{LoS}} \eta_{\mathrm{LoS}} + (1 - P_{\mathrm{LoS}}) \eta_{\mathrm{NLoS}}, \tag{2}$$

where η_{LoS} and η_{NLoS} denote respectively the additive loss incurred on top of the free space pathloss for line-of-sight (LoS) and non-line-of-sight (NLoS) links. f_c and c denote the carrier frequency and the speed of light, respectively. $d_{m,k}$ is the distance between the mth FAP and the kth user, which can be calculated by

$$d_{m,k} = \sqrt{(x_m - x_k)^2 + (y_m - y_k)^2 + {z_m}^2}, \tag{3}$$

P_{LoS} denotes the LoS probability of air-ground link, which can be calculated by

$$P_{\mathrm{LoS}} = \frac{1}{1 + a \exp\left(-b(\arctan(\frac{h}{r}) - a)\right)}, \tag{4}$$

where h and r denote the FAP flying altitude and the horizontal distance between the FAP and the ground user, respectively. (a, b) are environment-dependent variables.

B. Transmission Model

We consider the downlink transmission process, FAPs conduct beamforming to different users to reduce multiuser interference and improve spectrum efficiency. In this letter, to avoid sharing of channel state information between FAPs and reduce computational complexity, FAPs use conjugate beamforming to transmit signals to K users. In downlink, corresponding downlink signal received at the kth user is given by:

$$y_k = \sqrt{p_{\max}} \sum_{m=1}^{M} \eta_{m,k}^{1/2} g_{m,k} w_{m,k} q_k + \sqrt{p_{\max}} \sum_{m=1}^{M} \sum_{k' \neq k}^{K} \eta_{m,k'}^{1/2} g_{m,k} w_{m,k'} q_{k'} + n_k, \tag{5}$$

where $w_{m,k} = g_{m,k}^*$ is conjugate beamforming coefficients. q_k is the messages intended for the kth user, which satisfies $\mathbb{E}\{|q_k|^2\} = 1$. $p_{\max}$ and $\eta_{m,k}$ denote the total transmit power of each FAP and power allocation coefficients from mth FAP to kth user, respectively. $n_k \sim \mathcal{CN}(0, \sigma_0^2)$ is the additive white Gaussian noise (AWGN) and $\sigma_0^2 = BN_0$ is the noise power, N_0 is the power spectral density of noise, and B represents the channel bandwidth. The SINR of kth user can be given by

$$\gamma_k = \frac{p_{\max} \left| \sum_{m=1}^{M} g_{m,k} w_{m,k} \eta_{m,k}^{1/2} \right|^2}{\sigma_0^2 + p_{\max} \sum_{k' \neq k}^{K} \left| \sum_{m=1}^{M} g_{m,k} w_{m,k'} \eta_{m,k'}^{1/2} \right|^2} \tag{6}$$

Finally the achievable rate R_k for user k is given by

$$R_k = B \log_2(1 + \gamma_k). \tag{7}$$

C. Problem Formulation

In the considered downlink cell-free UAV network, our objective is to maximize the minimum user achievable rate by jointly adjusting the 3D location of FAPs and power allocation for the served users. The problem can be formulated as

$$\begin{aligned} \max_{\mathbf{q}_m, \eta_{m,k}} . \quad & \min_{k \in K} R_k \\ \text{s.t.} \quad & \text{C1}: \sum_{k=1}^{K} \eta_{m,k} \leq 1, \ \forall m \in \mathcal{M} \\ & \text{C2}: \eta_{m,k} \geq 0, \ \forall m \in \mathcal{M}, \forall k \in \mathcal{K} \\ & \text{C3}: d_{i,j} \geq d_{\min}, \ \forall i, j \in \mathcal{M}, i \neq j \end{aligned} \tag{8}$$

C1 represents in each FAP, the sum of the transmit power allocated to all users cannot exceed the maximum power. C2 ensures all users can be simultaneously served by all FAPs. C3 means the distance between any FAPs needs to be greater than the minimum safe distance $d_{\min}$.

III. Soft Actor-Critic Based 3D Deployment and Power Allocation Strategy

In the cell-free UAV network, FAPs are fully connected to users, which couples with 3D position of FAPs and power allocation coefficients, resulting in the joint optimization problem having high dimensions and computational complexity. Solving this problem with traditional DRL algorithms can be extremely challenging. For example, DQN cannot handle high-dimensional problems due to the discrete action space, and DDPG is easily trapped in a local optimum in the face of complex problems because of its weak exploration ability. In addition, DDPG has substantial hyperparameters that need to be adjusted, and their performance is sensitive to them. In contrast, SAC has fewer sensitive hyperparameters, which increases its robustness and stability. Therefore, SAC is more suitable for practical implementation in our safeguarding anti-collision cell-free UAV networks.

In this section, SAC is adopted to solve the joint 3D deployment and power allocation optimization. We first transform the original problem into a constrained Markov Decision Process (CMDP), including designing the specific state, action, and reward. Then the centralized SAC algorithm is utilized where the central controller acts as the agent and controls the behaviors of the FAPs. In contrast to other off-policy DRL algorithms, SAC is far more stable while inheriting the advantage of the high sampling efficiency of off-policy.

A. CMDP Formulation

We first transform the joint optimization problem into a CMDP problem. A CMDP can be represented by a tuple $(\mathcal{S}, \mathcal{A}, P, R, C)$, where $\mathcal{S}$ and $\mathcal{A}$ denote the global state space and action space, respectively, $P : \mathcal{S} \times \mathcal{S} \times \mathcal{A} \rightarrow [0, \infty)$ is the state transition probability function representing the density of the next state $\mathbf{s}_{t+1} \in \mathcal{S}$ given current state $\mathbf{s}_t \in \mathcal{S}$ and action $\mathbf{a}_t \in \mathcal{A}$. After taking action, the agent gets reward $r(\mathbf{s}_t, \mathbf{a}_t) \in \mathbb{R}$ according to the reward function $R : \mathcal{S} \times \mathcal{A} \rightarrow \mathbb{R}$ and cost function $C : \mathcal{S} \times \mathcal{A} \rightarrow \mathbb{R}$. We denote the historical state-action trajectory under the policy π by ρ_π. The aim of DRL is to learn the optimal policy π which maximizes the cumulative expected rewards and can be represented as

$$\pi^* = \arg\max_{\pi} \sum\nolimits_t \mathbb{E}_{(\mathbf{s}_t, \mathbf{a}_t) \sim \rho_\pi} [r(\mathbf{s}_t, \mathbf{a}_t)], \tag{9}$$

where π^* represents the optimal policy. In our work, to maximize the minimum user achievable rate, under the context that all FAPs cooperate to serve users, a careful design of state, action, and reward is required, which is described in detail as follows.

1) State: As mentioned in Section II, the achievable rate is related to the locations of FAPs and power allocation coefficients. The state space can be formulated as

$$\mathbf{s}_t = \left\{ \mathbf{q}_1^{t-1}, \ldots, \mathbf{q}_M^{t-1}, \eta_{1,1}^{t-1}, \ldots, \eta_{M,K}^{t-1} \right\}, \tag{10}$$

which has ($3M + MK$) dimensions.

2) Action: In the considered cell-free UAV network, the action space consists of two parts: 3D deployment of multi-FAPs and power allocation coefficients, described as

$$\mathbf{a}_t = \left\{ \mathbf{v}_1^t, \ldots, \mathbf{v}_M^t, \eta_{1,1}^t, \ldots, \eta_{M,K}^t \right\}, \tag{11}$$

which has ($M + MK$) dimensions, where $\{\mathbf{v}_m\}_{m \in \mathcal{M}} = [\Delta x_m, \Delta y_m, \Delta z_m]$ denote the movement changes in the X-axis, Y-axis and Z-axis of all M FAPs. Each FAP can choose one of the following actions in each state $\mathbf{s}_t$: "forward", "backward", "right", "left", "up", "down" and "still". $\{\eta_{m,k}\}_{m \in \mathcal{M}, k \in \mathcal{K}}$ is the power allocation coefficient from all FAPs to all users. In order to enable the applicability of SAC to our hybird action space which consists both discrete and continuous action, we perform mapping to extend the standard SAC algorithm that is limited to generating continuous values. The detailed operations are described in Section IV.

3) Reward: According to the problem modeled in (8), we design the reward function as two parts. Firstly, as the objective of the problem is to maximize the minimum user rate, the reward needs to be positively related to the objective function, when action $\mathbf{a}_t$ is chosen under state $\mathbf{s}_t$, we have

$$r(t) = \min\{R_1, \ldots, R_K\}. \tag{12}$$

Then, to meet the collision prevention requirement between FAPs, we subtract a penalty term from the cost function when the distance between FAPs is less than the minimum safety distance.

$$c(t) = N_t, \tag{13}$$

where N_t is the number of FAPs that do not meet the minimum safe distance constraint. Accordingly, the rewards function is defined as

$$r(\mathbf{s}_t, \mathbf{a}_t) = \varphi_1 r(t) + \mu c(t), \tag{14}$$

where φ_1 and μ are the adjusting parameters of two parts.

During the procedure of the off-line training, to realize precoding for interference elimination, small-scale coefficients is utilized to calculate reward. In the actual model deployment phase, the well-trained agent chooses the action only depend on the location of FAPs and users, which avoids huge overhead of acquiring channel state information.

B. SAC for 3D Deployment and Power Allocation

Based on the CMDP problem defined above, we present the learning part of the proposed SAC-based joint 3D deployment and power allocation algorithm. Unlike traditional DRL methods that only maximize the cumulative expected rewards as expressed in (9), SAC aims to maximize expected cumulative rewards while maximizing expected entropy of policy over ρ_π. This means taking actions as randomly as possible while succeeding at the task, which is more conducive to finding the optimal solution. Thus the aim can be reformulated as

$$J(\pi) = \sum_{t=0}^{T} \mathbb{E}_{(\mathbf{s}_t, \mathbf{a}_t) \sim \rho_\pi} [r(\mathbf{s}_t, \mathbf{a}_t) + \alpha \mathcal{H}(\pi(\cdot \mid \mathbf{s}_t))], \tag{15}$$

where α is a temperature parameter which determines the relative importance of entropy relative to reward, and $\mathcal{H}(\pi(\cdot|\mathbf{s}_t))$ is the entropy which is calculated as $\mathcal{H}(\pi(\cdot|\mathbf{s}_t)) = -\log \pi(\cdot|\mathbf{s}_t)$.

In SAC-based algorithm [13], the policy $\pi_\phi(\mathbf{a}_t|\mathbf{s}_t)$ is generated by the actor network. Meanwhile, in order to evaluate the performance of the actor network, an critic network is used to approximate soft Q-function $Q_\theta(\mathbf{s}_t, \mathbf{a}_t)$, which denotes the accumulative expected rewards after taking the action $\mathbf{a}_t$ at the state $\mathbf{s}_t$ under the policy π. Here, ϕ and θ are the parameters of the actor network and the critic network, respectively.

For the critic network, we maintain two Q networks to avoid overestimation of soft Q-values, using the corresponding two target networks for stable training. The soft Q-function parameters can be trained by minimizing the loss function

$$L_Q(\theta) = \mathbb{E}_{(\mathbf{s}_t,\mathbf{a}_t)\sim\mathcal{B}}[\frac{1}{2}(Q_\theta(\mathbf{s}_t,\mathbf{a}_t) - (r(\mathbf{s}_t,\mathbf{a}_t) + \gamma Q_{\bar{\theta}}(\mathbf{s}_{t+1},\mathbf{a}_{t+1}) - \log \pi_\phi(\mathbf{a}_{t+1} \mid \mathbf{s}_{t+1})))^2], \quad (16)$$

where $\mathcal{B}$ represents replay memory buffer, $\bar{\theta}$ is the parameter of target Q network, and updated periodically as $\bar{\theta} \leftarrow \tau\theta + (1-\tau)\bar{\theta}$ with $\tau \ll 1$.

For the actor network, we minimize the loss function as follow to update policy parameters

$$L_\pi(\phi) = \mathbb{E}_{\mathbf{s}_t\sim\mathcal{B}}\big[\mathbb{E}_{\mathbf{a}_t\sim\pi_\phi}[\alpha \log(\pi_\phi(\mathbf{a}_t|\mathbf{s}_t)) - Q_\theta(\mathbf{s}_t,\mathbf{a}_t)]\big]. \quad (17)$$

Finally, the loss function adjusting the temperature parameter α which balances the exploration and exploitation is expressed as follow

$$L(\alpha) = \mathbb{E}_{\mathbf{a}_t\sim\pi_t}\big[-\alpha_t \log \pi_t(\mathbf{a}_t|\mathbf{s}_t) - \alpha_t\overline{\mathcal{H}}\big], \quad (18)$$

where $\overline{\mathcal{H}} = -|\mathcal{A}|_{dim}$.

The SAC-based algorithm is a two-stage process consisting of training and implementation. Considering the demand of safeguard [14] and efficient training [15], we apply offline method to train the model. The agent utilizes statistical channel state information generated by simulator to train the neural networks at each step. During the model implementation stage, the well-trained model can be deployed to real networks for real-time and low-complexity decision-making processes.

IV. Simulation Results

In this section, to evaluate our proposed SAC-based joint 3D deployment and power allocation algorithm in the cell-free UAV network, we compare it with popular DRL algorithms, e.g., DDPG and twin delayed deep deterministic policy gradient (TD3), and conventional cell-based UAV network where each user can only be served by one FAP [1].

In our simulation, we consider K users randomly distributed in a 300 m $\times$ 300 m area, and the height h of the FAPs is limited between 100 m and 200 m. Communication-related parameters are set as follows: B =1 MHz, f_c = 2.4 GHz, $N_0 = -174$ dBm/Hz, the total transmit power of FAP $p_{\max} = 20$ dBm. $(a, b, \eta_{\mathrm{LoS}}, \eta_{\mathrm{NLoS}}) = (4.88, 0.43, 0.1, 21)$ are radio frequency propagation parameters of suburban environment [12]. The minimum distance between each FAP $d_{\min} = 20$ m and the corresponding penalty term $\mu = 20$.

For the parameter setting of the SAC algorithm, in terms of the neural network architecture, we use a four-layer fully connected structure, including two hidden layers with 256 neurons. ReLU is adopted as activation function in the critic network. The output layer of the action network has $(M + MK)$ dimensions and uses tanh as the activation function to limit the output value to $(-1, 1)$. A necessary mapping is performed to extend SAC to fit hybrid action space. We transform the first M dimensions of the output into integers from 0 to 6 by numerical transformation, representing each FAP's seven movement directions. The post MK dimensions of the output are converted into power allocation factor $\eta_{m,k} \in (0, 1)$. In addition, softmax function is utilized for normalization when the sum of $\eta_{m,k}$ in specific FAP exceeds the maximum power constraint.

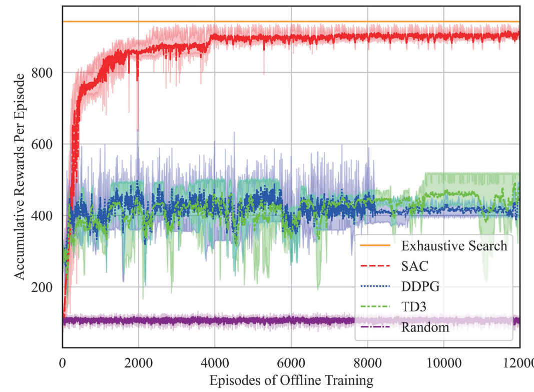

Fig. 2. Average rewards of different algorithms. Here $K = 2$, $M = 4$.

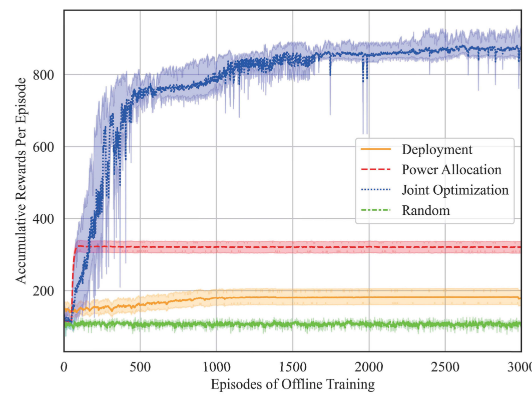

Fig. 3. Average rewards of different optimization dimensions. Here $K = 2$, $M = 4$.

The discount factor $\gamma = 0.95$, learning rate of neural networks is 3×10^{-4}, target smoothing coefficient $\tau = 5 \times 10^{-3}$.

The learning curves of the proposed SAC-based algorithm and other conventional algorithms are shown in Fig. 2, which demonstrated the excellent performance of SAC in solving complex optimization problems. It is evident that the SAC-based algorithm can converge close to result of exhaustive search, which is considered optimal benchmark. In addition, compared to TD3 and DDPG, SAC-based algorithm has faster convergence, greater stability, and better utility function optimization results, because it adds policy entropy to the optimization function while DDPG and TD3 increase exploration ability only by adding noise. All the DRL-based methods outperform the random policy, which confirms the effectivity of DRL.

As shown in Fig. 3, we compare the performance of the proposed joint 3D deployment and power allocation algorithm with fix deployment algorithm, fix power allocation algorithm, and random action algorithm. It can be seen that after the convergence, the average accumulative rewards of the joint optimization are much greater than just optimizing power allocation or location deployment. At the same time, they both outperform the random action algorithm, which proves the effectiveness of the optimization algorithm. In addition, the joint optimization algorithm requires a much larger state-action space to be explored than the single-dimensional optimization, therefore it converges more slowly.

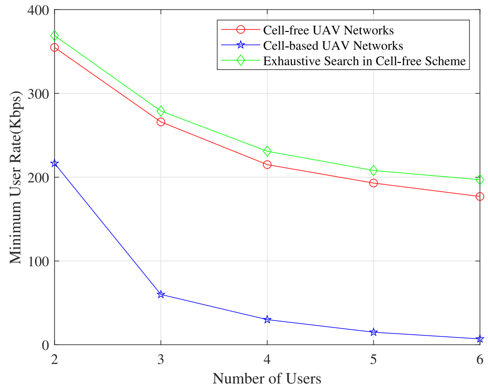

Fig. 4. Minimum user rate for different number of users. Here $M = 6$.

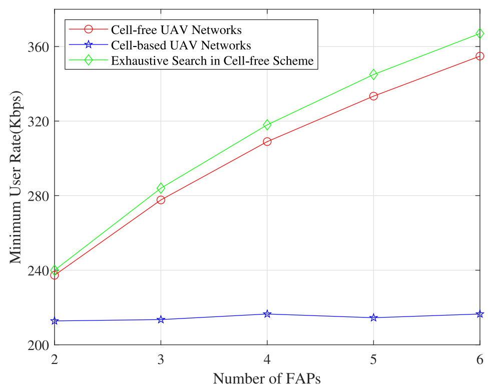

Fig. 5. Minimum user rate for different number of FAPs. Here $K = 2$.

Next, in Fig. 5 and Fig. 4, we compare the cell-free UAV network and the conventional cell-based network. In Fig. 5, to evaluate the effect of the number of FAPs, we fix the number of users as $K = 2$. In the cell-free scenario, where there is a fully connected relationship between FAPs and users, owing to the array gain, the performance of systems increases hugely when M increases. As a contrast, in cell-based UAV scenario, the number of FAPs increases with the little performance gain because FAPs and users have a one-to-one connection relationship. Furthermore, Fig. 4 shows the minimum user rate in different numbers of users, where $M = 6$. Due to the increasing interference, the user rate decreases as the number of users grows. As expected, the cell-free UAV network outperforms the cell-based UAV network because it benefits a lot from beamforming and suffers less from interference than the cell-based UAV network.

V. Conclusion

In this letter, we have studied the joint FAPs 3D deployment and power allocation for maximizing the minimum achievable user rate in the cell-free UAV network. We have proposed the SAC-based algorithm, which considers a more efficient maximum entropy objective and a fully centralized framework to allow FAPs to learn an optimal policy in high-dimensional state and action space, and alleviate the non-stationarity. Simulation results demonstrate that the cell-free UAV network can achieve a higher minimum user rate than conventional cell-based UAV network, and our proposed SAC-based algorithm outperforms conventional DRL-based algorithms in terms of stability and minimum user rate. In our work, DRL model needs to be retrained when the amount of user or FAP is changed. In the future, we will investigate a scalable DRL algorithm that incorporates possible techniques such as attention mechanism and meta-learning to better match the scalability of the cell-free UAV network where the number of FAPs and users can dynamically change.

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